Talinolol PBPK model: fit PK

1 Methods: Talinolol PBPK model: fit PK

1.1 Parameters and data

8 parameters were optimized: tissue/plasma partition coefficient tal (`Kp_tal`), tissue distribution tal (`ftissue_tal`), renal clearance of unbound talinolol relative to the GFR (`KI__TALEX_f`), Vmax for talinolol import (`LI__TALIM_Vmax`), Ka_dis [1/hr] dissolution tal (`Ka_dis_tal`), rate of the apical uptake of talinolol (`GU__TALABS_k`), Vmax of the efflux of talinolol by P-glycoprotein (`GU__TALEFL_Vmax`) and Km of the efflux of talinolol by P-glycoprotein (`GU__TALEFL_Km`).

A fit mapping is a curve of a study, the data and the observable of the model it is compared with. The fit has 85 fit mappings of training data with 874 data points, whose residuals enter the objective; 64 fit mappings of validation data with 756 data points, which were simulated and evaluated with the fitted parameters but not fitted; and 12 fit mappings of outliers with 90 data points, data the fit dropped as not usable, which were evaluated like the validation data so that the decision to drop them can be checked. The fit was set up and run with sbmlsim [1].

1.2 Objective function

The parameters were estimated by minimizing the objective function

$$F(\vec{p}) = \frac{1}{2} \sum_k \sum_i \rho\left(w_{i,k} r_{i,k}(\vec{p})^2\right) \tag{1}$$

where k runs over the fit mappings of the training data and i over their data points, $r_{i,k}$ is the residual, $w_{i,k} = w_k w^p_{i,k}$ the weight of the data point and ρ the loss function, applied to the squared weighted residual $z = w_{i,k} r^2_{i,k}$. The loss function is linear, $\rho(z) = z$, i.e. a standard least squares problem.

The residuals are normalized by the mean $\overline{y}_k$ of the data of their curve, which makes curves of different quantities and magnitudes comparable:

$$r_{i,k} = \frac{y_{i,k} - f_{k(x_{i,k},\vec{p})}}{\overline{y}_k} \tag{2}$$

The weight of curve k is $w_k = \frac{n_k}{N_k}$, the weight n_k of its fit mapping (the weight its fit mapping collection assigns, 1 by default, or, for a collection which uses the weights of its mappings, the weight of the mapping, by default the number of subjects of its data) divided by the number N_k of its data points.

A data point is weighted with the inverse of its coefficient of variation, $w^p_{i,k} = |y_{i,k} \frac{|}{\sigma_{i,k}}$, with $\sigma_{i,k}$ the standard deviation of the data point, or its standard error for data without a standard deviation. An error of zero is missing, a missing error of a curve with errors is replaced by the largest error of the curve, and the data points of a curve without errors have the weight $w^p_{i,k} = 2$, the weight of a coefficient of variation of 0.5.

1.3 Optimization

The optimizer searched the parameters in log10 space within their bounds; bounds, values and intervals are given in the units of the fit parameters, the units of the table of the parameters.

The objective was minimized in 3 runs with differential evolution [2] of SciPy [3] within the bounds of the parameters, with the strategy `best1bin`, a population of 15 times the number of parameters, at most 1000 generations, the relative tolerance 0.01 of convergence, the recombination constant 0.7 and a mutation dithered between 0.5 and 1. The initial population of a run was drawn by latin hypercube sampling [4], with a seed of every run derived from the seed 1234 of the fit. The best member of the final population was polished with the L-BFGS-B method of SciPy.

The convergence of the runs was assessed from the traces of their cost and the waterfall plot of their final costs, and the parameter set with the lowest cost was used.

The models were simulated with libRoadRunner [5] and its integrator CVODE [6] with the relative tolerance 10^{-6} and the absolute tolerances 10^{-12} of an amount, 10^{-12} of a concentration and 10^{-6} of the other states, which for a species in a concentration is multiplied by the initial volume of its compartment, and 1 state of its own. It ran with the variable step size of libRoadRunner, i.e. with an output at every step of the integrator. A data point at the time of a change after the start of a simulation, e.g. a dose or the reset of a collection, was compared with the state before the change, except the first data point of a curve with data at later times, which starts a collection and was compared with the state after the change.

1.4 Goodness of fit

The goodness of fit is given per kind of data (training data, validation data and outliers) by metrics over its n data points with the residuals $e_i = y_i - f\left(x_i, \hat{\vec{p}}\right)$ at the fitted parameters $\hat{\vec{p}}$, the mean squared error, its root and the root mean squared error normalized by the mean $\overline{y}_{k(i)}$ of the data of the curve of a point:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} e_i^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \quad \mathrm{NRMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{\overline{y}_{k(i)}}\right)^2} \tag{3}$$

The root mean square of the residuals of the objective, i.e. what the fit minimizes, and the coefficient of determination:

$$\mathrm{RMSE}_w = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\rho(w_i r_i^2)}, \quad R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i \left(y_i - \overline{y}\right)^2} \tag{4}$$

The predictions of a non-linear model are no linear regression of the data, so R^2 can be negative: the prediction is worse than the mean of the data. The information criteria of Akaike [7] and Schwarz [8] with the number n_p of fitted parameters are those of a least squares fit with normally distributed residuals, up to an additive constant, so only differences between models fitted to the same data are meaningful:

$$\mathrm{AIC} = n\ln(\mathrm{MSE}) + 2n_p, \quad \mathrm{BIC} = n\ln(\mathrm{MSE}) + n_p\ln(n) \tag{5}$$

The percentage of the data points whose prediction $f_i = f\left(x_i, \hat{\vec{p}}\right)$ lies within twofold of the data is taken over the n_+ data points with positive data $y_i > 0$ and a positive prediction $f_i > 0$; the others have no ratio and are not counted, and a kind of data without such a point has no percentage:

$$P_2 = \frac{100}{n_+}\left|\left\{i : \frac{1}{2} \le \frac{f_i}{y_i} \le 2\right\}\right| \tag{6}$$

The goodness of fit plot shows the predictions against the data on logarithmic axes, with the lines of twofold $f = 2y$ and $f = \frac{y}{2}$. The Bland-Altman plot [9] shows the ratio of prediction and data as $d_i = \log_{10}\left(\frac{f_i}{y_i}\right)$ against the geometric mean $\sqrt{f_i y_i}$ for the data points with positive values, with the bias $\overline{d}$ and the limits of agreement $\overline{d} \pm 1.96 s_d$, with the sample standard deviation s_d of d_i, of the training data; they are read as the fold factors $10^{\overline{d}}$ and $10^{\overline{d} \pm 1.96 s_d}$.

1.5 Identifiability

The identifiability of the parameters was analysed with the profile likelihood [10], [11]. The profile of a parameter is the lowest objective with the parameter fixed:

$$\mathrm{PL}(p_j) = \min_{p_{l \ne j}} F(\vec{p}) \tag{7}$$

At every point of a scan the other parameters were re-optimized from the previous point with the least squares optimizer of SciPy with the relative step 0.05. A scan ran from the optimum in both directions in the space of the optimizer, with steps in decades of the parameter, starting at 0.1. A step which raised the objective by more than 0.2 of the distance between the objective at the optimum and the threshold was reduced by the factor 2 and repeated, down to 0.01, and the step after one which raised it by less than that divided by the factor was enlarged by the factor, up to 1. A scan stopped at the threshold, at the bound of the parameter, at a point which failed or after 50 points.

The confidence intervals are the values at which the profile crosses the threshold of the likelihood ratio test

$$F_{\mathrm{thr}} = F_{\mathrm{min}} + \chi^2_\alpha \frac{\mathrm{df}}{2} \tag{8}$$

with F_{min} the lowest objective of the fit and of all profiles and $\chi^2_\alpha(\mathrm{df})$ the quantile of the χ^2 distribution at the confidence level $\alpha = 0.95$ with df = 1 degrees of freedom, i.e. 1.92 above the lowest objective. The threshold has this confidence level if F is the negative log-likelihood of the data up to a constant, i.e. if the weighted residuals are independent and normally distributed with unit variance; otherwise the level is approximate. A crossing is interpolated between two points of the profile, on the logarithm of a parameter on a logarithmic scale. A parameter is identifiable if its profile reaches the threshold within the scan on both sides of the optimum, i.e. its confidence interval is finite; practically non-identifiable towards the lower bound if its profile reaches the threshold above the optimum but not below it within the scan, which ends at the lower bound or after 50 points; practically non-

identifiable towards the upper bound if its profile reaches the threshold below the optimum but not above it within the scan, which ends at the upper bound or after 50 points; practically non-identifiable if its profile reaches the threshold on neither side within the scans, which end at the bounds or after 50 points, and is not flat; and structurally non-identifiable if its profile is flat on both sides. A side of a profile is flat if it rises by less than 0.05 of the distance between the lowest objective and the threshold, and a profile with a point which failed or did not converge is not classified. The profile likelihood is preferred over the intervals of the Fisher information, its intervals are invariant under a transformation of the parameters and can be asymmetric [12].

The Fisher information is the local alternative to the profiles: it is the Gauss-Newton approximation of the curvature of the objective at the optimum, from the Jacobian J of the weighted residuals of the training data with respect to the parameters $\vec{p}$ in the space of the optimizer, computed by central finite differences with the step $h_j = \varepsilon \max\left(|p_j|, 1\right),\ \varepsilon = 10^{-6}$ of the parameter p_j:

$$I = J^{\top} J, \quad \mathrm{cov} = \sigma^2 I^{-1}, \quad \sigma^2 = \frac{2F}{n - n_p} \tag{9}$$

with the number n of data points of the training data and the number n_p of parameters. The information has rank 8 of 8, the number of its eigenvalues above 10^{-8} of the largest one, and the condition number 1950; the covariance of an information of a lower rank is computed with the pseudo-inverse. The Wald intervals are $\hat{p}_j \pm t_{\frac{1+\alpha}{2}, n-n_p} \mathrm{SE}_j$ with the standard errors $\mathrm{SE}_j = \sqrt{\mathrm{cov}_{jj}}$ at the confidence level $\alpha = 0.95$, transformed back into the units of the fit parameters.

1.6 Provenance

The fit PK_20261020_042735__49b48 was run on 2026-10-20 02:27:38 UTC with sbmlsim 0.10.0 [1], libRoadRunner 2.10.0 [5], SciPy 1.18.1 [3], NumPy 2.5.3 [13] and Python 3.14.4 (seed 1234). The models and data were those of commit `9597cdc55eeaa70a613cfdd136415a468c8583f8` of `/home/mkoenig/git/pkdb_models-talinolol` their SHA-256 checksums are listed in Table 1.

Table 1: SHA-256 checksums of the models and data of the fit PK_20261020_042735__49b48, their first 12 characters; the full checksums are in the provenance of the result (optimization_result.json).

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_Fig2.tsv	bf75c1259ea0...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_Fig4a.tsv	8193e464127d...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_Fig4b.tsv	6402c1c03afd...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_Fig5.tsv	5ad611b3d609...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_Tab1.tsv	8952b50ad229...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_Tab1a.tsv	f43b60f6d158...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_Tab2.tsv	473ea57df4aa...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_TabGroups.tsv	26bd6133a94a...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_TabIndividuals.tsv	60d41879d828...
pkdb_models/models/talinolol/data/talinolol/Bernsdorf2006/.Bernsdorf2006_TabInterventions.tsv	f50cd3d88ac5...
pkdb_models/models/talinolol/data/talinolol/Bogman2005/.Bogman2005_Fig1.tsv	058a772a6901...
pkdb_models/models/talinolol/data/talinolol/Bogman2005/.Bogman2005_Fig2.tsv	27c068f1d379...
pkdb_models/models/talinolol/data/talinolol/Bogman2005/.Bogman2005_Tab2.tsv	a94801d8cb46...
pkdb_models/models/talinolol/data/talinolol/Bogman2005/.Bogman2005_TabGroups.tsv	28ba0dd810e0...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Bogman2005/.Bogman2005_TabIndividuals.tsv	e724d81e5cbe...
pkdb_models/models/talinolol/data/talinolol/Bogman2005/.Bogman2005_TabInterventions.tsv	010cba44ee94...
pkdb_models/models/talinolol/data/talinolol/DeMey1995/.DeMey1995_Fig4.tsv	0867b203550b...
pkdb_models/models/talinolol/data/talinolol/DeMey1995/.DeMey1995_Fig4a.tsv	8136c8905644...
pkdb_models/models/talinolol/data/talinolol/DeMey1995/.DeMey1995_TabGroups.tsv	570d9dc1cb46...
pkdb_models/models/talinolol/data/talinolol/DeMey1995/.DeMey1995_TabInterventions.tsv	74613b74bec4...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_Fig2.tsv	a988af771670...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_Fig3.tsv	aad89a9ee9b6...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_Fig4.tsv	e3800353846e...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_Fig5.tsv	52ea8f2a16be...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_TabData.tsv	23c729c54e91...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_TabGroups.tsv	56c0fcde9e08...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_TabIndividuals.tsv	9adeedd54038...
pkdb_models/models/talinolol/data/talinolol/Drozdzik2014/.Drozdzik2014_TabInterventions.tsv	33937b898d35...
pkdb_models/models/talinolol/data/talinolol/Fan2009a/.Fan2009a_Fig1.tsv	44d348cddee1...
pkdb_models/models/talinolol/data/talinolol/Fan2009a/.Fan2009a_Fig2.tsv	2f831b757354...
pkdb_models/models/talinolol/data/talinolol/Fan2009a/.Fan2009a_Tab1.tsv	b3e0c1a7453a...
pkdb_models/models/talinolol/data/talinolol/Fan2009a/.Fan2009a_TabGroups.tsv	47a9f98e2258...
pkdb_models/models/talinolol/data/talinolol/Fan2009a/.Fan2009a_TabIndividuals.tsv	633e50fc924f...
pkdb_models/models/talinolol/data/talinolol/Fan2009a/.Fan2009a_TabInterventions.tsv	556d6ddd2f3b...
pkdb_models/models/talinolol/data/talinolol/Fan2009b/.Fan2009b_Fig1.tsv	6b5473c1e7a3...
pkdb_models/models/talinolol/data/talinolol/Fan2009b/.Fan2009b_Fig2.tsv	56c873b7ec3f...
pkdb_models/models/talinolol/data/talinolol/Fan2009b/.Fan2009b_Tab1.tsv	b68d93d07e11...
pkdb_models/models/talinolol/data/talinolol/Fan2009b/.Fan2009b_TabGroups.tsv	ff68c316bfda...
pkdb_models/models/talinolol/data/talinolol/Fan2009b/.Fan2009b_TabIndividuals.tsv	fffc1f777809...
pkdb_models/models/talinolol/data/talinolol/Fan2009b/.Fan2009b_TabInterventions.tsv	db00982b3f60...
pkdb_models/models/talinolol/data/talinolol/Giessmann2004/.Giessmann2004_Fig2.tsv	c31a48396d38...
pkdb_models/models/talinolol/data/talinolol/Giessmann2004/.Giessmann2004_Tab1.tsv	620a16cc4ede...
pkdb_models/models/talinolol/data/talinolol/Giessmann2004/.Giessmann2004_Tab3.tsv	0754a2941be2...
pkdb_models/models/talinolol/data/talinolol/Giessmann2004/.Giessmann2004_Tab3a.tsv	efbe2506c618...
pkdb_models/models/talinolol/data/talinolol/Giessmann2004/.Giessmann2004_TabGroups.tsv	b4ac874c8113...
pkdb_models/models/talinolol/data/talinolol/Giessmann2004/.Giessmann2004_TabInterventions.tsv	f946ca729626...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Gramatte1996/.Gramatte1996_Fig1.tsv	7148f529eef6...
pkdb_models/models/talinolol/data/talinolol/Gramatte1996/.Gramatte1996_Fig3.tsv	32a9f7b7b4a1...
pkdb_models/models/talinolol/data/talinolol/Gramatte1996/.Gramatte1996_Tab1.tsv	a7297c2719b0...
pkdb_models/models/talinolol/data/talinolol/Gramatte1996/.Gramatte1996_TabData.tsv	1fc49627f782...
pkdb_models/models/talinolol/data/talinolol/Gramatte1996/.Gramatte1996_TabGroups.tsv	1eff4dcef2a9...
pkdb_models/models/talinolol/data/talinolol/Gramatte1996/.Gramatte1996_TabIndividuals.tsv	e602f181a22c...
pkdb_models/models/talinolol/data/talinolol/Gramatte1996/.Gramatte1996_TabInterventions.tsv	a31358dc7a41...
pkdb_models/models/talinolol/data/talinolol/Han2009/.Han2009_Fig1.tsv	e73de06bb13e...
pkdb_models/models/talinolol/data/talinolol/Han2009/.Han2009_Fig2.tsv	f4d77f971bb4...
pkdb_models/models/talinolol/data/talinolol/Han2009/.Han2009_Tab1.tsv	832764c2140f...
pkdb_models/models/talinolol/data/talinolol/Han2009/.Han2009_Tab2.tsv	76f148fca9cb...
pkdb_models/models/talinolol/data/talinolol/Han2009/.Han2009_TabGroups.tsv	5039be4b61df...
pkdb_models/models/talinolol/data/talinolol/Han2009/.Han2009_TabIndividuals.tsv	52cdf8ea5a83...
pkdb_models/models/talinolol/data/talinolol/Han2009/.Han2009_TabInterventions.tsv	16ccba1e0cb4...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Fig1.tsv	e4347de5b689...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Fig2.tsv	6ea054ea23ea...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab1.tsv	ba5f0fb98481...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab1_mean.tsv	c30e381dd516...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab1a.tsv	16de55e1411e...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab1a_mean.tsv	5225f8dc2f2a...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab2.tsv	dcc8c44b96e5...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab2a.tsv	5f31e6d1aa92...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab3.tsv	4ab226bb2876...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_Tab3a.tsv	964485aa3b1c...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_TabGroups.tsv	1f65bdeb9aa3...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_TabIndividuals.tsv	64417e0b2d8a...
pkdb_models/models/talinolol/data/talinolol/Haustein1981/.Haustein1981_TabInterventions.tsv	e62b9fb2b508...
pkdb_models/models/talinolol/data/talinolol/He2007/.He2007_Fig2.tsv	5e2563a92e37...
pkdb_models/models/talinolol/data/talinolol/He2007/.He2007_Tab4.tsv	29574293d4b3...
pkdb_models/models/talinolol/data/talinolol/He2007/.He2007_TabGroups.tsv	8b8b16ee3826...
pkdb_models/models/talinolol/data/talinolol/He2007/.He2007_TabInterventions.tsv	d6639ee5c55e...
pkdb_models/models/talinolol/data/talinolol/He2012/.He2012_Fig1.tsv	b3130c435ad0...
pkdb_models/models/talinolol/data/talinolol/He2012/.He2012_Fig2.tsv	e4cc32b9f2db...
pkdb_models/models/talinolol/data/talinolol/He2012/.He2012_Fig3.tsv	667067c794c4...
pkdb_models/models/talinolol/data/talinolol/He2012/.He2012_Tab1.tsv	2011d1fae876...
pkdb_models/models/talinolol/data/talinolol/He2012/.He2012_Tab2.tsv	d74e9e03f596...
pkdb_models/models/talinolol/data/talinolol/He2012/.He2012_TabGroups.tsv	85d9571301e6...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/He2012/.He2012_TabInterventions.tsv	3cbc944d2431...
pkdb_models/models/talinolol/data/talinolol/Juan2007/.Juan2007_Fig2.tsv	c0c5937dde28...
pkdb_models/models/talinolol/data/talinolol/Juan2007/.Juan2007_Tab1.tsv	c7b46f3ae6c3...
pkdb_models/models/talinolol/data/talinolol/Juan2007/.Juan2007_TabGroups.tsv	3d9b1645a8fe...
pkdb_models/models/talinolol/data/talinolol/Juan2007/.Juan2007_TabInterventions.tsv	afba60d3dfdc...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Fig1.tsv	0910acf458b3...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Fig2.tsv	dfc153f12433...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Fig3.tsv	6c42959da848...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Fig3a.tsv	9c5d33bcada4...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Tab1.tsv	4a0ff4df8110...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Tab1a.tsv	b7e078531fe6...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Tab2.tsv	b0073d72988e...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Tab2a.tsv	080f8eb7f58c...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_Tab2b.tsv	e2d3a7cbb645...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_TabData.tsv	0178925db668...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_TabGroups.tsv	2e1861fded6c...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_TabIndividuals.tsv	1a1740a3d8dd...
pkdb_models/models/talinolol/data/talinolol/Krueger2001/.Krueger2001_TabInterventions.tsv	2f7b90ca79cd...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_Fig1.tsv	18d61766cc8a...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_Fig2.tsv	3af12c8b12d5...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_Fig3.tsv	c3b6336517b9...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_Fig4.tsv	542d12896b6d...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_Tab2.tsv	1563445e2c7f...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_Tab2a.tsv	7a5562014b2a...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_Tab3.tsv	f726e236a541...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_TabData.tsv	fd9a76143704...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_TabGroups.tsv	65d763f9d0ef...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_TabIndividuals.tsv	c45fe2ce2f8e...
pkdb_models/models/talinolol/data/talinolol/Nguyen2014/.Nguyen2014_TabInterventions.tsv	12be79f258dc...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_Fig1.tsv	f48e4d0435ce...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_Fig2.tsv	f7eb804c2dbf...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_Fig3.tsv	f7d40861321b...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_Fig3c.tsv	429abc7b1613...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_Tab1.tsv	7bfcb40ec3c4...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_Tab1a.tsv	a6a78135d123...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_TabData.tsv	f40aef646671...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_TabGroups.tsv	544881170216...
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_TabIndividuals.tsv	0af397c0555c...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Nguyen2015/.Nguyen2015_TabInterventions.tsv	3dbe00ba800b...
pkdb_models/models/talinolol/data/talinolol/Ruike2010/.Ruike2010_Fig3.tsv	08fa540693db...
pkdb_models/models/talinolol/data/talinolol/Ruike2010/.Ruike2010_Tab1.tsv	7bfcf16796fa...
pkdb_models/models/talinolol/data/talinolol/Ruike2010/.Ruike2010_TabGroups.tsv	383713c2f061...
pkdb_models/models/talinolol/data/talinolol/Ruike2010/.Ruike2010_TabInterventions.tsv	bb24ca25ff40...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_Fig1.tsv	28f3d18f2a32...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_Tab1.tsv	c2d1a4c1d837...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_Tab1_mean.tsv	0580d47252da...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_Tab1a.tsv	fe1818527427...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_Tab1a_mean.tsv	10f9f3555c0b...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_Tab2.tsv	6c42ad4149c6...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_Tab2_mean.tsv	212f04abd844...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_TabGroups.tsv	41b8c67eee65...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_TabIndividuals.tsv	df934f7454de...
pkdb_models/models/talinolol/data/talinolol/Schwarz1999/.Schwarz1999_TabInterventions.tsv	4be73e73390f...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_Fig1.tsv	cc6a57146e4a...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_Tab1.tsv	5f8de57423a1...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_Tab1_mean.tsv	3a76380d3096...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_Tab1a.tsv	8fcd5a6324fa...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_Tab1a_mean.tsv	47847d52ddda...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_TabGroups.tsv	2ff2978dac5b...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_TabIndividuals.tsv	dc4e24fa3c14...
pkdb_models/models/talinolol/data/talinolol/Schwarz2000/.Schwarz2000_TabInterventions.tsv	cc55eb1218a6...
pkdb_models/models/talinolol/data/talinolol/Schwarz2005/.Schwarz2005_Fig1.tsv	3b09195bfd29...
pkdb_models/models/talinolol/data/talinolol/Schwarz2005/.Schwarz2005_Fig2.tsv	049d2c906f1a...
pkdb_models/models/talinolol/data/talinolol/Schwarz2005/.Schwarz2005_Tab1.tsv	dfa370bf8c7b...
pkdb_models/models/talinolol/data/talinolol/Schwarz2005/.Schwarz2005_Tab1a.tsv	3f4737e1697c...
pkdb_models/models/talinolol/data/talinolol/Schwarz2005/.Schwarz2005_Tab2.tsv	b141c4622aab...
pkdb_models/models/talinolol/data/talinolol/Schwarz2005/.Schwarz2005_TabGroups.tsv	e1c2bf0bf031...
pkdb_models/models/talinolol/data/talinolol/Schwarz2005/.Schwarz2005_TabInterventions.tsv	781bd42abc83...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Fig1AB.tsv	d3d0294fadec...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Fig1CD.tsv	d9edac0c435e...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Fig2mRNA.tsv	111e2e1050f4...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Fig3A.tsv	57b74edbec16...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Fig3mRNA.tsv	2029667358f1...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Tab1.tsv	7d5cb895cdc9...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Tab1a.tsv	927321752c33...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Tab3.tsv	b80c4fc4039e...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_Tab3mRNA.tsv	e4b7cebd8ae2...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_TabGroups.tsv	a1cf56096456...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_TabIndividuals.tsv	f034552a54d3...
pkdb_models/models/talinolol/data/talinolol/Schwarz2007/.Schwarz2007_TabInterventions.tsv	93a65f60e007...
pkdb_models/models/talinolol/data/talinolol/Siegmund2003/.Siegmund2003_Fig1.tsv	4c853c51abd6...
pkdb_models/models/talinolol/data/talinolol/Siegmund2003/.Siegmund2003_Tab1.tsv	da62d45cddc4...
pkdb_models/models/talinolol/data/talinolol/Siegmund2003/.Siegmund2003_TabGroups.tsv	f7ad181d0d33...
pkdb_models/models/talinolol/data/talinolol/Siegmund2003/.Siegmund2003_TabInterventions.tsv	f5498d15cc9c...
pkdb_models/models/talinolol/data/talinolol/Terhaag1983/.Terhaag1983_Fig1.tsv	fb19bfef5fc6...
pkdb_models/models/talinolol/data/talinolol/Terhaag1983/.Terhaag1983_Tab1.tsv	dfec22ddd2e6...
pkdb_models/models/talinolol/data/talinolol/Terhaag1983/.Terhaag1983_TabGroups.tsv	6afa4992bc64...
pkdb_models/models/talinolol/data/talinolol/Terhaag1983/.Terhaag1983_TabIndividuals.tsv	b97217e6defb...
pkdb_models/models/talinolol/data/talinolol/Terhaag1983/.Terhaag1983_TabInterventions.tsv	6313c63adc9a...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_Fig1.tsv	ead0b4b006ef...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_Tab1.tsv	2e0a9c629e6a...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_Tab1_median.tsv	76659dece9ef...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_Tab1a.tsv	3553c76c7ca0...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_Tab1a_mean.tsv	18cc53494142...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_Tab1median.tsv	76659dece9ef...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_TabGroups.tsv	9addeea6eb2f...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_TabIndividuals.tsv	482f0b6d74d9...
pkdb_models/models/talinolol/data/talinolol/Terhaag1989/.Terhaag1989_TabInterventions.tsv	8d3759cfe2b1...
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_Fig1.tsv	c8412141f80f...
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_Fig2.tsv	ff3284dbec52...
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_Fig3.tsv	1a89229dfc53...
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_Tab1.tsv	dc21e44c829a...
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_Tab1a.tsv	d7b3f557723d...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_TabGroups.tsv	07e1cda45669...
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_TabIndividuals.tsv	3c9ed8817987...
pkdb_models/models/talinolol/data/talinolol/Trausch1995/.Trausch1995_TabInterventions.tsv	fdce4ef9384e...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_Fig3.tsv	73658e58dd1b...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_Fig4.tsv	490bda06ff19...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_Fig5.tsv	9ed2cc370d3d...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_Tab1.tsv	b7fc074c48a3...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_Tab1a.tsv	93689284faa2...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_TabGroups.tsv	07330b8904de...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_TabIndividuals.tsv	912af1032412...
pkdb_models/models/talinolol/data/talinolol/Tubic2006a/.Tubic2006a_TabInterventions.tsv	47da1acf4a4e...
pkdb_models/models/talinolol/data/talinolol/Wang2013/.Wang2013_Fig2.tsv	d43e8ddfeaea...
pkdb_models/models/talinolol/data/talinolol/Wang2013/.Wang2013_Tab1.tsv	e9bf7081751f...
pkdb_models/models/talinolol/data/talinolol/Wang2013/.Wang2013_Tab3.tsv	045e14a073d7...
pkdb_models/models/talinolol/data/talinolol/Wang2013/.Wang2013_TabGroups.tsv	78439f740daa...
pkdb_models/models/talinolol/data/talinolol/Wang2013/.Wang2013_TabInterventions.tsv	9fd8c0cdf9a3...
pkdb_models/models/talinolol/data/talinolol/Westphal2000a/.Westphal2000a_Fig1.tsv	44df550518ae...
pkdb_models/models/talinolol/data/talinolol/Westphal2000a/.Westphal2000a_Fig2.tsv	0bdbab50decf...
pkdb_models/models/talinolol/data/talinolol/Westphal2000a/.Westphal2000a_Tab1.tsv	ce3795a94ca9...
pkdb_models/models/talinolol/data/talinolol/Westphal2000a/.Westphal2000a_Tab2.tsv	47f59339f8b5...
pkdb_models/models/talinolol/data/talinolol/Westphal2000a/.Westphal2000a_Tab2a.tsv	d37fa228933b...
pkdb_models/models/talinolol/data/talinolol/Westphal2000a/.Westphal2000a_TabGroups.tsv	bfa49dcba4d8...
pkdb_models/models/talinolol/data/talinolol/Westphal2000a/.Westphal2000a_TabInterventions.tsv	5df7a83454bf...
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_Fig2.tsv	5232910880be...
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_Fig3.tsv	f4038ef20b24...
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_Fig4.tsv	f4038ef20b24...
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_Tab1.tsv	1f693ef9781c...
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_Tab1a.tsv	06eb3b178638...
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_Tab2.tsv	d0ba9014959d...
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_TabGroups.tsv	6879b435284c...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Westphal2000b/.Westphal2000b_TabInterventions.tsv	5ea3a558f339...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_Fig1.tsv	d063fc2dd720...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_Fig2.tsv	35fb55eea4aa...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_Fig3.tsv	af443c7a0c2b...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_Tab1a.tsv	1899ce6af39b...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_Tab1b.tsv	4202b0c9e323...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_Tab1c.tsv	ab425da2fbdc...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_TabGroups.tsv	9cc0d990218c...
pkdb_models/models/talinolol/data/talinolol/Wetterich1996/.Wetterich1996_TabInterventions.tsv	546f3590bfb9...
pkdb_models/models/talinolol/data/talinolol/Xiao2012/.Xiao2012_Fig1.tsv	37332d57bf12...
pkdb_models/models/talinolol/data/talinolol/Xiao2012/.Xiao2012_Fig2.tsv	88f8f136d12e...
pkdb_models/models/talinolol/data/talinolol/Xiao2012/.Xiao2012_Fig3.tsv	4dedd7dc8dd0...
pkdb_models/models/talinolol/data/talinolol/Xiao2012/.Xiao2012_Tab1.tsv	649018cb5493...
pkdb_models/models/talinolol/data/talinolol/Xiao2012/.Xiao2012_Tab2.tsv	7d4b21aca8f1...
pkdb_models/models/talinolol/data/talinolol/Xiao2012/.Xiao2012_TabGroups.tsv	76bf16af8ac1...
pkdb_models/models/talinolol/data/talinolol/Xiao2012/.Xiao2012_TabInterventions.tsv	637f193f7a08...
pkdb_models/models/talinolol/data/talinolol/Yan2013/.Yan2013_Fig2.tsv	e96a31fe99ae...
pkdb_models/models/talinolol/data/talinolol/Yan2013/.Yan2013_Fig3.tsv	1cc9bde3dd80...
pkdb_models/models/talinolol/data/talinolol/Yan2013/.Yan2013_Tab2.tsv	4df2e6a265c7...
pkdb_models/models/talinolol/data/talinolol/Yan2013/.Yan2013_TabGroups.tsv	d76550d2248e...
pkdb_models/models/talinolol/data/talinolol/Yan2013/.Yan2013_TabIndividuals.tsv	c48697897ecd...
pkdb_models/models/talinolol/data/talinolol/Yan2013/.Yan2013_TabInterventions.tsv	d73315649f7a...
pkdb_models/models/talinolol/data/talinolol/Zeng2009/.Zeng2009_Fig1.tsv	4cb7f14b5112...
pkdb_models/models/talinolol/data/talinolol/Zeng2009/.Zeng2009_Fig2.tsv	b417f559c4e9...
pkdb_models/models/talinolol/data/talinolol/Zeng2009/.Zeng2009_Tab1.tsv	0d426d593be4...
pkdb_models/models/talinolol/data/talinolol/Zeng2009/.Zeng2009_TabGroups.tsv	15f4552c7e8f...
pkdb_models/models/talinolol/data/talinolol/Zeng2009/.Zeng2009_TabIndividuals.tsv	c3bb66305236...
pkdb_models/models/talinolol/data/talinolol/Zeng2009/.Zeng2009_TabInterventions.tsv	3381d0b286c5...
pkdb_models/models/talinolol/data/talinolol/Zhang2005/.Zhang2005_Fig1.tsv	4b73956cd7af...
pkdb_models/models/talinolol/data/talinolol/Zhang2005/.Zhang2005_Tab4.tsv	d56646704503...
pkdb_models/models/talinolol/data/talinolol/Zhang2005/.Zhang2005_TabGroups.tsv	e66309094703...
pkdb_models/models/talinolol/data/talinolol/Zhang2005/.Zhang2005_TabInterventions.tsv	5c939063bb71...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_Fig4.tsv	ab3a354acddc...

File	SHA-256
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_Fig5.tsv	dcdc1d061673...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_Tab1.tsv	05189b4c10b2...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_Tab2.tsv	d7459a27c2cf...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_Tab3.tsv	bc5d469c743f...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_Tab4.tsv	915f1b05a336...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_TabGroups.tsv	4e459bea8552...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_TabIndividuals.tsv	4ae946166ef9...
pkdb_models/models/talinolol/data/talinolol/Zschiesche2002/.Zschiesche2002_TabInterventions.tsv	fb32cee9da41...
pkdb_models/models/talinolol/models/output/talinolol_body_flat.xml	eb9dd592c078...

2 Results: Talinolol PBPK model: fit PK

2.1 Parameters

Table 2: Parameters of the fit PK_20261020_042735__49b48, the parameter set PK_20261020_042735__49b48_0, cost 132, with their bounds in the optimization. The 95% confidence intervals are those of the profile likelihood, LB and UB an interval which is open towards the lower or the upper bound, and the Wald intervals of the Fisher information for a parameter without a profile. Identifiable is the classification of the profile: yes, no (LB) or no (UB) if the profile does not reach the threshold towards the lower or the upper bound, no if it reaches it on neither side, structural if it is flat.

Parameter	Description	Value [95% CI]	Unit	Bounds	Identifiable
Kp_tal	tissue/plasma partition coefficient tal	3.92 [3.11, 4.92]	dimensionless	[0.3, 30]	yes
ftissue_tal	tissue distribution tal	2.76 [1.97, 5.46]	l/min	[0.01, 50]	yes
KI__TALEX_f	renal clearance of unbound talinolol relative to the GFR	7.66 [6.64, 9.71]	dimensionless	[1, 20]	yes
LI__TALIM_Vmax	Vmax for talinolol import	0.0369 [0.0226, 0.0847]	mmol/min/l	[0.0001, 1]	yes
Ka_dis_tal	Ka_dis [1/hr] dissolution tal	1.41 [1.03, 2.25]	1/hr	[0.1, 10]	yes
GU__TALABS_k	rate of the apical uptake of talinolol	0.00358 [0.00281, 0.00484]	1/min	[0.0001, 1]	yes
GU__TALEFL_Vmax	Vmax of the efflux of talinolol by P-glycoprotein	0.000103 [3.46e-05, 0.000289]	mmol/min/l	[1e-05, 1]	yes
GU__TALEFL_Km	Km of the efflux of talinolol by P-glycoprotein	0.00024 [5.76e-05, 0.000634]	mmol/l	[1e-05, 1]	yes

2.2 Data

Table 3: Data of the fit PK_20261020_042735__49b48: the number of fit mappings and, in parentheses, of data points per study and kind. The training data enters the cost, the validation data and the outliers are evaluated only.

Study	Training	Validation	Outlier	Total
Bernsdorf2006	6 (47)	4 (47)	-	10 (94)
Bogman2005	-	3 (42)	-	3 (42)
DeMey1995	4 (32)	-	-	4 (32)
Fan2009a	1 (14)	2 (28)	-	3 (42)
Fan2009b	1 (14)	2 (28)	-	3 (42)
Giessmann2004	1 (2)	2 (17)	-	3 (19)
Gramatte1996	-	8 (151)	-	8 (151)
Han2009	1 (13)	-	-	1 (13)
Haustein1981	-	3 (15)	6 (37)	9 (52)
He2007	1 (14)	-	-	1 (14)
He2012	1 (14)	2 (28)	-	3 (42)
Krueger2001	2 (19)	7 (89)	-	9 (108)
Nguyen2015	2 (26)	-	-	2 (26)
Ruike2010	1 (16)	-	-	1 (16)
Schwarz1999	2 (18)	-	-	2 (18)
Schwarz2000	-	2 (18)	-	2 (18)
Schwarz2005	-	-	2 (19)	2 (19)
Schwarz2007	4 (45)	4 (45)	-	8 (90)
Siegmund2003	2 (36)	-	-	2 (36)
Terhaag1983	3 (43)	-	-	3 (43)
Terhaag1989	5 (51)	5 (54)	3 (21)	13 (126)
Trausch1995	32 (335)	-	-	32 (335)
Tubic2006a	2 (14)	6 (59)	-	8 (73)
Wang2013	-	-	1 (13)	1 (13)
Westphal2000a	4 (32)	4 (32)	-	8 (64)
Westphal2000b	2 (19)	6 (51)	-	8 (70)
Wetterich1996	5 (30)	-	-	5 (30)
Xiao2012	-	1 (13)	-	1 (13)
Yan2013	1 (15)	-	-	1 (15)
Zeng2009	1 (12)	-	-	1 (12)
Zhang2005	1 (13)	3 (39)	-	4 (52)
Total	85 (874)	64 (756)	12 (90)	161 (1720)

2.3 Goodness of fit

Table 4: Goodness of fit of the fit PK_20261020_042735__49b48 per kind of data, the parameter set PK_20261020_042735__49b48_0. The cost is defined on the training data only, 2-fold is the percentage of the data points with a positive value and prediction whose prediction is within twofold of the data.

Kind	n	cost	MSE	RMSE	NRMSE	RMSE_w	R^2	2-fold [%]	AIC	BIC
training	874	132	1.52e-05	0.0039	0.579	0.55	0.863	84.1	−9678.5	−9640.3
validation	756	-	0.000413	0.0203	4.74	2.92	−2.23	64.4	−5875.2	−5838.2
outlier	90	-	0.00032	0.0179	2.36	3.36	−6.11	21.7	−708.2	−688.2

2.4 Figures

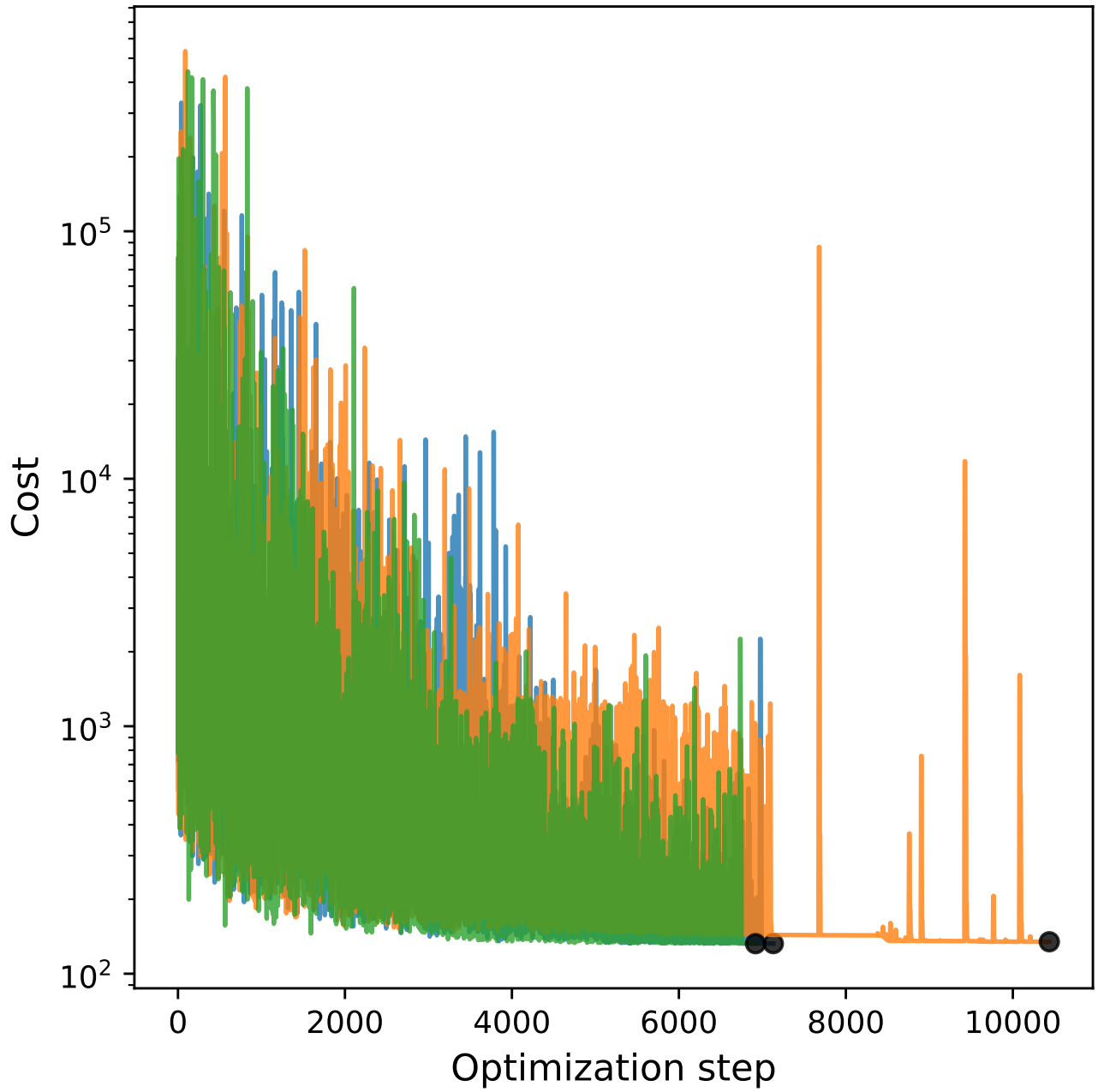


Figure 1: Cost of the optimizers over their steps, fit PK_20261020_042735__49b48. The cost of every optimization run over its steps. Runs which end at the same cost found the same optimum; a run which stops much higher is stuck in a local minimum.

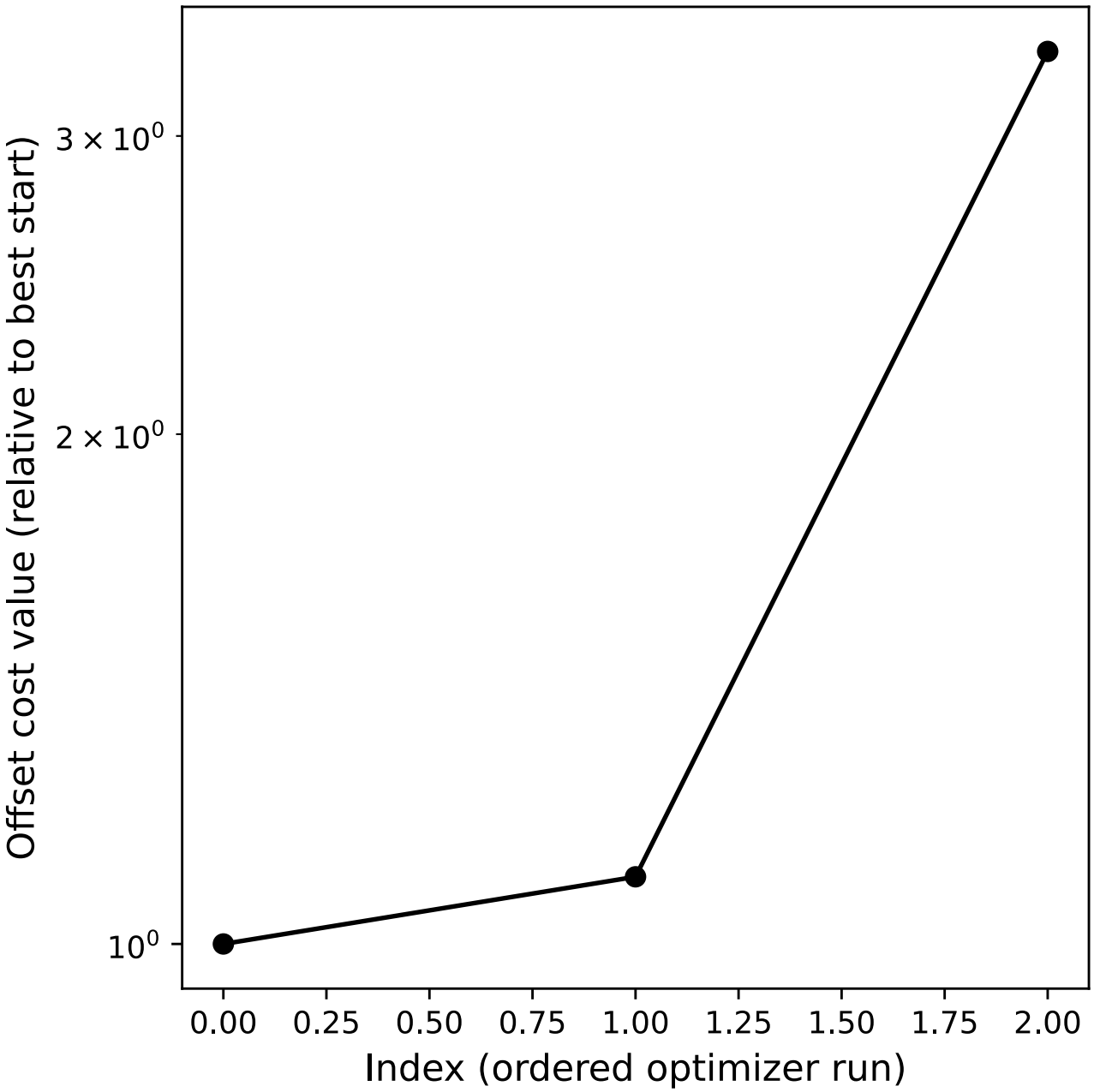


Figure 2: Cost of the optimization runs, ordered, fit PK_20261020_042735__49b48. The final cost of the runs, ordered. A flat plateau at the left is the global optimum found repeatedly, which is the evidence that the multistart converged; steps to the right are local minima.

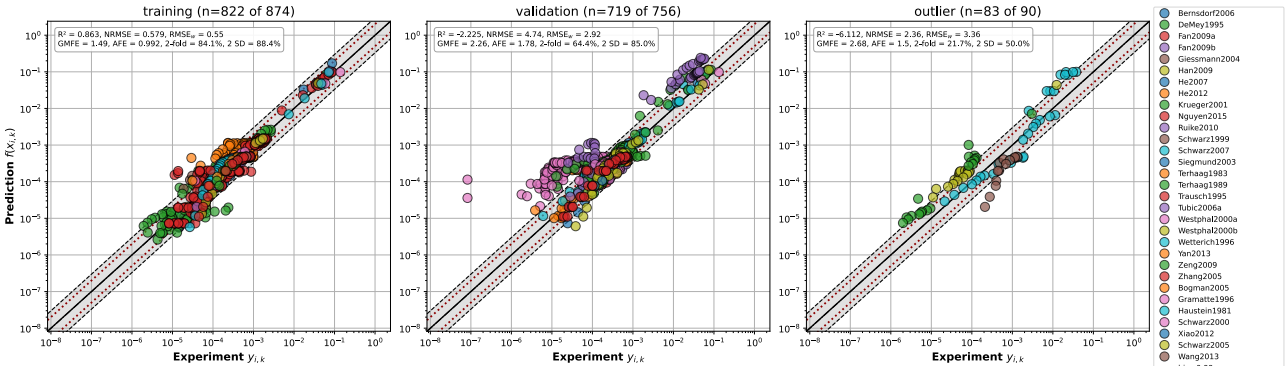


Figure 3: Prediction against the measured data points, per kind of fit mapping, with the agreement of the training data, the lines of twofold and the metrics of the panel, fit PK_20261020_042735__49b48. Within twofold of the data are 691 of the 822 training data points (84.1%), 463 of the 719 validation data points (64.4%) and 18 of the 83 outlier data points (21.7%); only the data points with a positive value and prediction are counted. Prediction against measurement, one point per data point, with a panel per kind of fit mapping. The points scatter around the diagonal when the model describes the data; a systematic deviation from it is a systematic error of the model. The band is the agreement of the training data, i.e. the same band the Bland-Altman plot draws, here as lines parallel to the diagonal. The dark red dotted lines are twofold, prediction = 2 x measurement and prediction = measurement / 2; the caption states the points between them of every kind. A point which is zero or negative is not drawn on the logarithmic axes and not counted, the title of a panel then states the points drawn of all. The validation panel shows how the fit describes data it was not fitted on, the outlier panel the data the fit dropped. The box of a panel carries its R^2, its normalized RMSE, the RMSE of the residuals of the cost, its fold errors GMFE and AFE and the percentages of its points within twofold and within two standard deviations of the data.

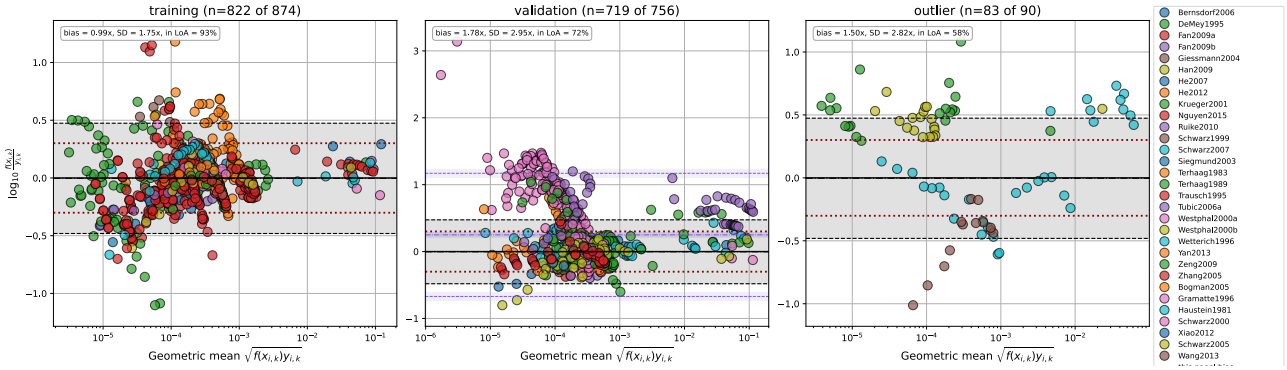


Figure 4: Agreement of prediction and measurement as a ratio, per kind of fit mapping, with the agreement of the training data and the agreement of the panel, fit PK_20261020_042735__49b48. The ratio of prediction and measurement over the geometric mean of the two, i.e. the goodness of fit with the diagonal turned into the horizontal. The band is the same, the bias and the limits of agreement `bias ± 1.96 SD` of the training data as fold factors. A bias away from 1 is a systematic over- or underprediction, wide limits are a large scatter, and a trend over the mean is a model which describes the large or the small values better. The box of a panel carries the bias and the SD of its own points as fold factors and the share of them inside the limits of agreement of the training data.

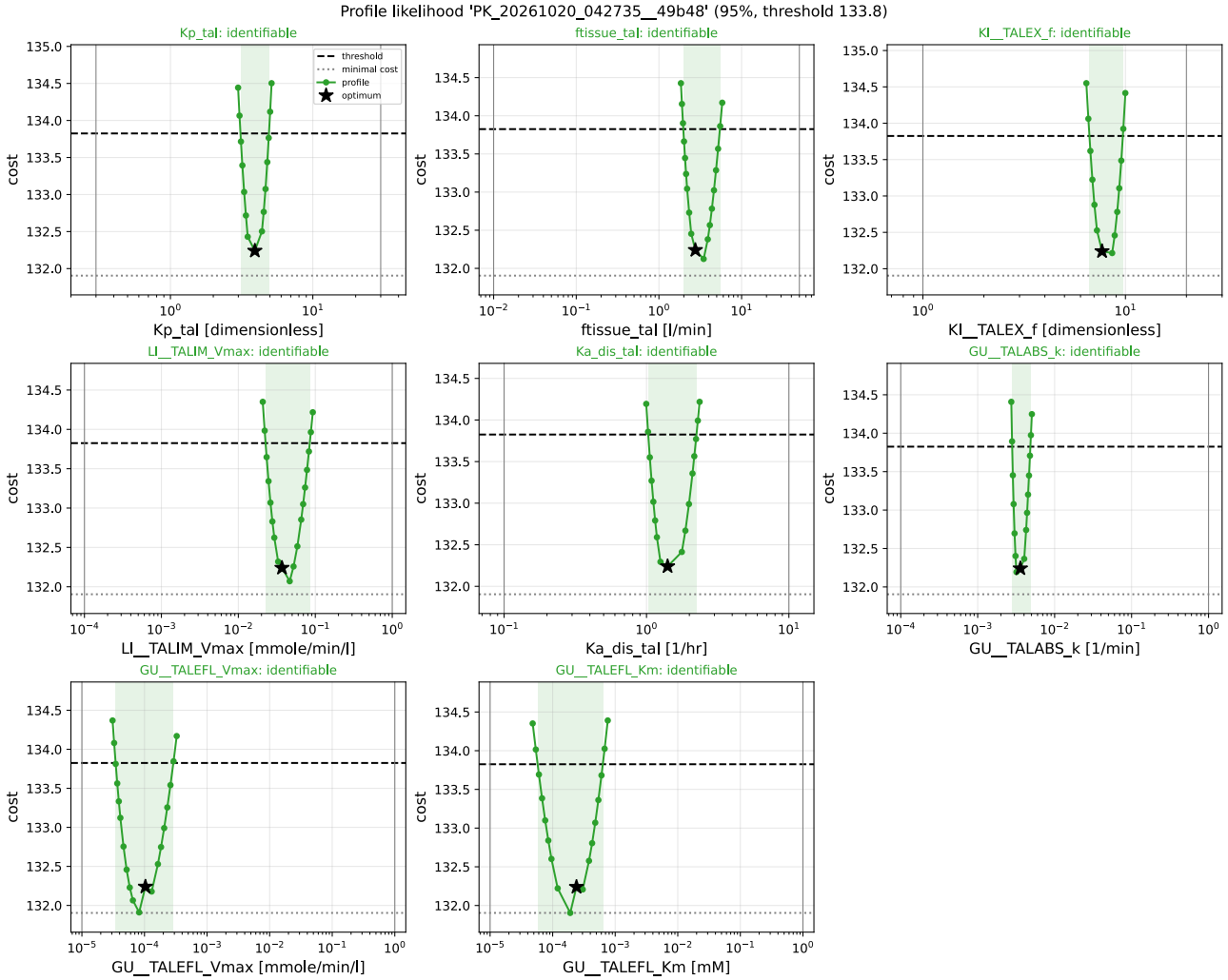


Figure 5: Profile likelihood of every parameter: the cost with the parameter fixed and the other parameters optimized, the threshold of the confidence level and the confidence interval, fit PK_20261020_042735__49b48. One profile per parameter. A profile which crosses the dashed threshold on both sides gives a finite confidence interval; one which stays below it up to a bound is practically non-identifiable; a flat profile is structurally non-identifiable, the parameter is compensated by the others.

2.5 Provenance

Table 5: Provenance of the fit PK_20261020_042735__49b48: the versions it ran with and the revision of its models and data; the SHA-256 checksums of the models and data are listed in Table 1.

Item	Value
Fit	`PK_20261020_042735__49b48`
Created	2026-10-20 02:27:38 UTC
Seed	1234
Python	3.14.4
sbmlsim	0.10.0
libRoadRunner	2.10.0
SciPy	1.18.1
NumPy	2.5.3
Repository	`/home/mkoenig/git/pkdb_models-talinolol`
Commit	`9597cdc55eeaa70a613cfdd136415a468c8583f8`
Uncommitted changes	no

Bibliography

[1] M. König, "sbmlsim: SBML simulation made easy." Zenodo, 2026. doi: 10.5281/zenodo.3597770. Version 0.10.0.

[2] R. Storn and K. Price, "Differential evolution - a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997, doi: 10.1023/A:1008202821328.

[3] P. Virtanen *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020, doi: 10.1038/s41592-019-0686-2. Version 1.18.1.

[4] M. D. McKay, R. J. Beckman, and W. J. Conover, "A comparison of three methods for selecting values of input variables in the analysis of output from a computer code," *Technometrics*, vol. 21, no. 2, pp. 239–245, 1979, doi: 10.2307/1268522.

[5] C. Welsh, J. Xu, L. Smith, M. König, K. Choi, and H. M. Sauro, "libRoadRunner 2.0: a high performance SBML simulation and analysis library," *Bioinformatics*, vol. 39, no. 1, p. btac770, 2023, doi: 10.1093/bioinformatics/btac770. Version 2.10.0.

[6] A. C. Hindmarsh *et al.*, "SUNDIALS: suite of nonlinear and differential/algebraic equation solvers," *ACM Transactions on Mathematical Software*, vol. 31, no. 3, pp. 363–396, 2005, doi: 10.1145/1089014.1089020.

[7] H. Akaike, "A new look at the statistical model identification," *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, 1974, doi: 10.1109/TAC.1974.1100705.

[8] G. Schwarz, "Estimating the dimension of a model," *The Annals of Statistics*, vol. 6, no. 2, pp. 461–464, 1978, doi: 10.1214/aos/1176344136.

[9] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986, doi: 10.1016/S0140-6736(86)90837-8.

[10] A. Raue *et al.*, "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009, doi: 10.1093/bioinformatics/btp358.

[11] C. Kreutz, A. Raue, D. Kaschek, and J. Timmer, "Profile likelihood in systems biology," *The FEBS Journal*, vol. 280, no. 11, pp. 2564–2571, 2013, doi: 10.1111/febs.12276.

[12] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, and J. Timmer, "On structural and practical identifiability," *Current Opinion in Systems Biology*, vol. 25, pp. 60–69, 2021, doi: 10.1016/j.coisb.2021.03.005.

[13] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020, doi: 10.1038/s41586-020-2649-2. Version 2.5.3.